

- **Option B: Pure NoSQL (MongoDB):** Store everything as documents.
- **Option C:** Use PostgreSQL for core profiles/billing and MongoDB for activity logs.
- **Chosen option:** Option A: Pure relational (PostgreSQL) with JSONB

Justification for the choice

Although Option C is theoretically feasible, the two distinct backup strategies, two security strategies, and the enormous effort of synchronisation between them can lead to significant operational complexity.

On the other hand, option A's PostgreSQL offers ACID compliance by default, and its JSONB capability enables the system to store unstructured activity logs with MongoDB-like performance. In a single, sophisticated database engine, this offers the best of both worlds. For storing and working with JSON (JavaScript Object Notation) data, PostgreSQL has a very effective binary data type called JSONB. JSONB's binary format makes it significantly faster for processing and querying compared to the plain-text JSON type.

The positives of this choice are:

- **Simplified operations:** One database technology for scaling, backup, and monitoring.
- **Strong integrity:** The system can guarantee the integrity of activity logs in relation to a valid user ID by using the user ID as a foreign key.
- **SQL power:** The software team can use standard BI tools (Tableau/Metabase).

The downsides are:

- **Storage size:** Large PostgreSQL indexes on JSONB may necessitate a stringent data retention strategy by restricting the amount of time logs that are stored.
- **Vertical scaling limits:** In contrast to MongoDB, Postgres scaling is primarily vertical (larger servers) as opposed to horizontal (sharding), which could result in additional costs.

The golden rule for a software architect is to consider the design that is easy to implement both in terms of human and operational resources (hardware and software). It is preferable to view a design as immutable. For any significant modification, it is advisable to make a new design and designate the previous one as 'superseded' rather than altering the original. This helps us learn from earlier mistakes and maintains the history of the system's evolution. **END** 🐧

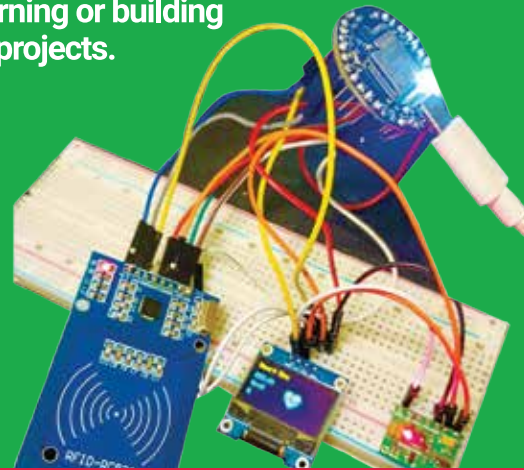
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